

OSI Layer

Q1. OSI 7-layer model with an example at each layer

OSI 7-Layer Model		
Layer	Example device / protocol	What it does
7 Application	HTTP / HTTPS, DNS (protocols)	User-facing services: browser, apps, web pages
6 Presentation	TLS / SSL, JPEG (encryption, formats)	Encryption, compression, data format conversion
5 Session	NetBIOS, RPC (session control)	Opens, manages and closes sessions
4 Transport	TCP, UDP (protocols, ports)	End-to-end delivery, segments, flow control
3 Network	Router, IP (device and protocol)	Logical (IP) addressing and routing of packets
2 Data Link	Switch, Ethernet (device, MAC address)	Frames, MAC addressing, error detection
1 Physical	Cables, Hub, WiFi radio (hardware)	Sends raw bits as electrical, light or radio signals

Figure 1: OSI model (layer 7 at the top, layer 1 at the bottom). Memory trick from layer 1 to 7: Please Do Not Throw Sausage Pizza Away.

Q2. Matching scenarios to OSI layers

Scenario	Layer Name (Layer Number)
a) Encrypting data before sending	Presentation (Layer 6)
b) Routing packets across networks	Network (Layer 3)
c) Converting data to electrical signals	Physical (Layer 1)
d) Opening a web page in your browser	Application (Layer 7)
e) Detecting and correcting errors in frames	Data Link (Layer 2)

Answers in the required format: a) Encrypting data: Presentation (6); b) Routing packets: Network (3); c) Converting data to electrical signals: Physical (1); d) Opening a web page: Application (7); e) Detecting and correcting errors in frames: Data Link (2).

Q3. OSI layers involved when I send a message on WhatsApp

When I type a message and press send, the data moves down the layers on my phone. On the receiver's phone it moves up the same layers in reverse order.

Layer	What happens in WhatsApp
Application (7)	The WhatsApp app takes my typed text or photo and hands it to the network as a message to be sent.
Presentation (6)	The message is end-to-end encrypted and media such as photos is compressed, so only the receiver can read it.
Transport (4)	TCP splits the data into segments, adds port numbers, and makes sure every segment arrives in the correct order.
Network (3)	IP adds my phone's IP address and the server's IP address. Routers use these to move the packets across the Internet.
Data Link (2)	The packets are placed in frames with the MAC addresses of my phone and the WiFi router for the next hop.
Physical (1)	The frames are turned into radio waves (WiFi) or mobile signals, then into light or electrical signals on cables.

Q4. Cannot open a website, but ping to its IP address works

Layers that are fine: A successful ping (ICMP) proves that **Physical (1), Data Link (2) and Network (3)** are working. My device is connected, it can reach the router, and IP routing to that server works.

Layers that may be the problem: the fault is **above the Network layer**, most likely in these layers:

Layer	Possible cause
Application (7)	Most likely. DNS may not be converting the website name to an IP address, the browser or proxy settings may be wrong, or the web server (HTTP/HTTPS service) itself may be down or returning errors.
Transport (4)	A firewall may be blocking the web ports (80 for HTTP or 443 for HTTPS), even though ping is allowed, so TCP connections fail.
Presentation (6)	A TLS/SSL problem, such as an expired certificate or wrong date and time, can stop a secure (HTTPS) site from loading.

Conclusion: Because ping works, the problem is not in layers 1 to 3. The best place to start is the Application layer (check DNS, try another browser) and then the Transport layer (check firewall and ports 80/443).